

2. Algebraic Expressions

Transformation of Algebraic Expressions

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Ratios and Rates

A **ratio** is a comparison of two quantities by division.

The ratio of a to b can be written in three different ways: a to b, $a:b$, and $\frac{a}{b}$.

If two quantities are in the ratio of a to b, then the two numbers can be expressed as ax and bx, in which x is a positive integer.

A **rate** is a ratio of two measurements having different units of measure. For example, a price of \$2.59 per gallon of gasoline, an income of \$750 in 3 days, and a speed of 60 miles per hour are all rates.

A **unit rate** is a rate that has a denominator of 1. Some examples of unit rates are defined as follows.

$$\textbf{Unit Price} = \frac{\text{Price of Package}}{\text{Number of Units in the Package}}$$

$$\textbf{Gas Mileage} = \frac{\text{Number of Miles Traveled}}{\text{Number of Gallons of Gas Used}}$$

$$\textbf{Speed (Miles per Hour)} = \frac{\text{Number of Miles Traveled}}{\text{Number of Hours}}$$

$$\textbf{Density} = \frac{\text{Mass}}{\text{Volume}}$$

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Example 1 □ Express each ratio as a unit rate.

a. 1360 grams of coffee cost $17.68. What is the unit price of the coffee?

b. A car travels 322 miles on 11.5 gallons of gas. What is the car's gas mileage?

c. A driver traveled $485\frac{1}{3}$ miles in $8\frac{2}{3}$ hours. What is his speed?

d. A volume of $46\ cm^3$ of silver has a mass of 483 grams. What is the density of silver?

Example 2 □ 3 angles of a triangle are in the ratio of $3:5:7$. What is the measure of each angle

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EXERCISES - 1

The ratio of $1\frac{3}{4}$ to $2\frac{1}{2}$ is equal to the ratio of 14 to what number?

A) 18

B) 20

C) 22

D) 24

EXERCISES - 2

The sum of two numbers is 14 and the ratio of the two numbers is -3. What is the product of the two numbers?

A) -105

B) -119

C) -133

D) -147

EXERCISES - 3

If $2(x-y)=3y$, what is the ratio $\frac{x}{y}$?

A) $\frac{2}{5}$

B) $\frac{4}{3}$

C) $\frac{5}{2}$

D) $\frac{8}{3}$

EXERCISES - 4

The ratio of length to width of a rectangular garden is $6:7$. If the perimeter of the rectangle is 78 meters, what is the area of the garden in square meters?

A) 274

B) 326

C) 352

D) 378

EXERCISES - 5

A car travels 218.5 miles on 9.5 gallons of gas.
What is the car's gas mileage?

Proportional

A **proportion** is an equation stating that two ratios are equal.

The proportions $\frac{a}{b}=\frac{c}{d}$ is read "a is to b as c is to d." The numbers a and d are called the **extremes** of the proportion. The numbers b and c are called the **means** of the proportion.

In a proportion, the product of the extremes is equal to the product of the means.

If $\frac{a}{b}=\frac{c}{d}$, then $ad=bc$. The products ad and bc are called the **cross products** of the proportion $\frac{a}{b}=\frac{c}{d}$.

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Example 1 □ Determine whether each pair of ratios forms a proporti

a. $\frac{0.4}{1.5}, \frac{1.6}{6}$

b. $\frac{12}{25}, 15$

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Example 2 □ Solve the proportion $\frac{3}{7} = \frac{6}{x-4}$.

Example 3 □ Carter's SUV requires 8 gallons of gasoline to travel 148 miles. How much gasoline, to the nearest gallon, will he need for a 500 mile trip?

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